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Report on

"Simulation Of 4 Bit PIPO Shift Register"

Submitted by

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Experiment: Simulate 4-bit PIPO shift register (parallel in, parallel out).

Aim:

To Simulate 4-bit PIPO shift register (parallel in, parallel out). Input parallel data "1011" and show parallel output instantly after the active clock edge.

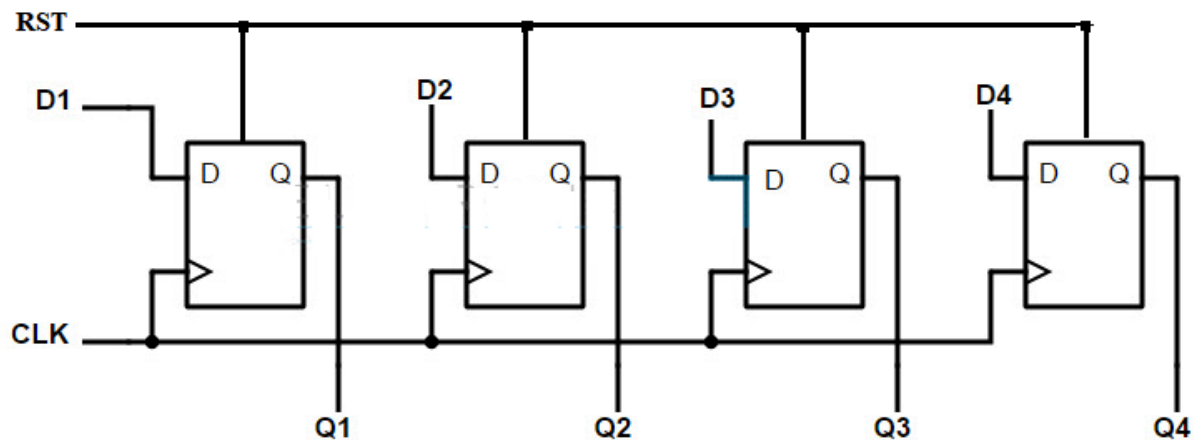
Components Required:

- D Flip-Flops - 4
- Clock Pulse Generator - 1
- Toggle Switches (Parallel Inputs) – 4
- LED Indicators - 4
- Clear/Reset Line - 1
- Connecting Wires

Theory:

A 4-bit Parallel-In Parallel-Out (PIPO) Shift Register is a sequential logic circuit that allows data to be loaded simultaneously across all inputs and read simultaneously across all outputs. Unlike a Serial-In Serial-Out (SISO) or Serial-In Parallel-Out (SIPO) shift register, the PIPO configuration does not shift bits from one flip-flop to the next. Instead, each flip-flop independently captures the data present at its respective D input on the triggering edge of a common clock pulse.

The circuit consists of four D Flip-Flops connected in parallel. The data inputs D0, D1, D2, and D3 are directly connected to the inputs of the flip-flops. A common clock line (CLK) is distributed to all flip-flops to ensure synchronous data loading. Additionally, a clear/reset line is provided to initialize all outputs (Q0, Q1, Q2, Q3) to 0000. Upon receiving a clock pulse, the entire 4-bit word is instantly stored and made available at the outputs, enabling high-speed parallel data transfer without any shifting delays.



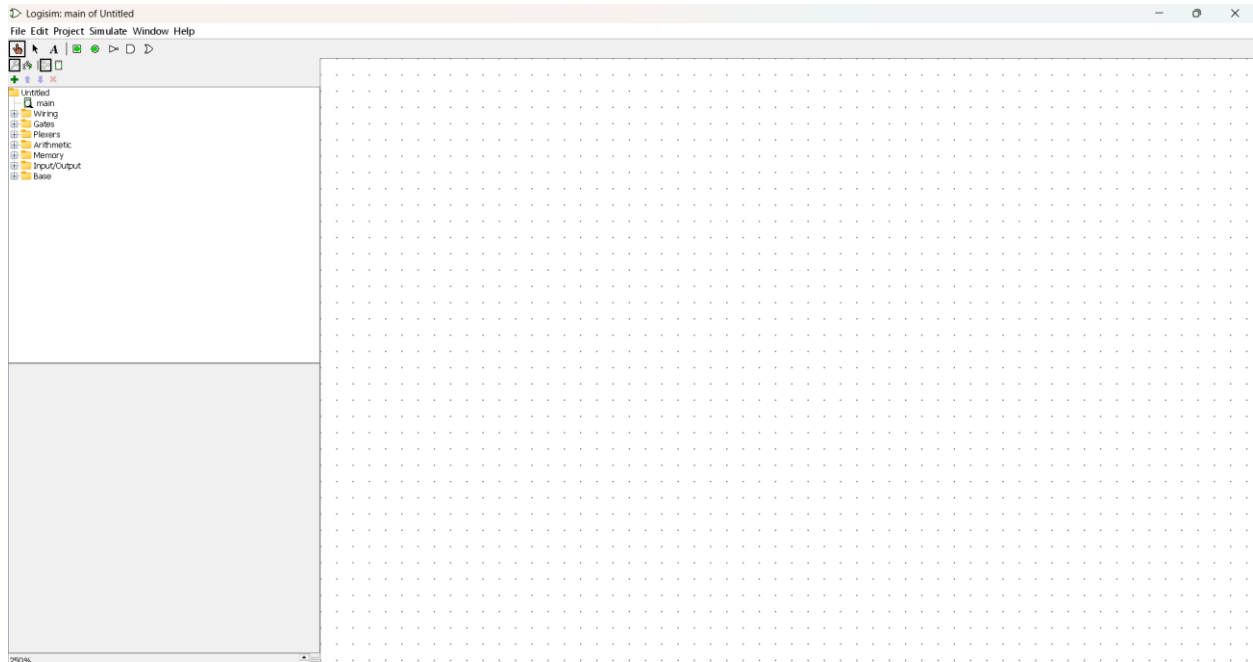
Block Diagram - 4 Bit PIPO shift Register

[Image Placeholder: Neat labeled block diagram of a 4-bit PIPO Shift Register showing D0-D3 parallel inputs, Common Clock, Clear, and Q0-Q3 parallel outputs]

A 4-bit PIPO Shift Register contains:

- Four Parallel Inputs (D0, D1, D2, D3)
- One Common Clock Input (CLK)
- One Clear/Reset Input
- Four D Flip-Flops
- Four Parallel Outputs (Q0, Q1, Q2, Q3)

The parallel data is loaded simultaneously across all flip-flops and appears at the parallel outputs immediately upon the active edge of the common clock pulse.



Truth Table / Output Table of 4-bit PIPO Shift Register

(Parallel input = 1011)

Clock Pulse	Parallel Inputs (D3 D2 D1 D0)	Parallel Outputs (Q3 Q2 Q1 Q0)
Initial (Before clock)	1 0 1 1	0 0 0 0
Clock 1	1 0 1 1	1 0 1 1
Clock 2	0 1 1 0	0 1 1 0
Clock 3	1 1 1 1	1 1 1 1
Clock 4	0 0 0 0	0 0 0 0

(Q3 is the leftmost flip-flop output and Q0 is the rightmost flip-flop output)

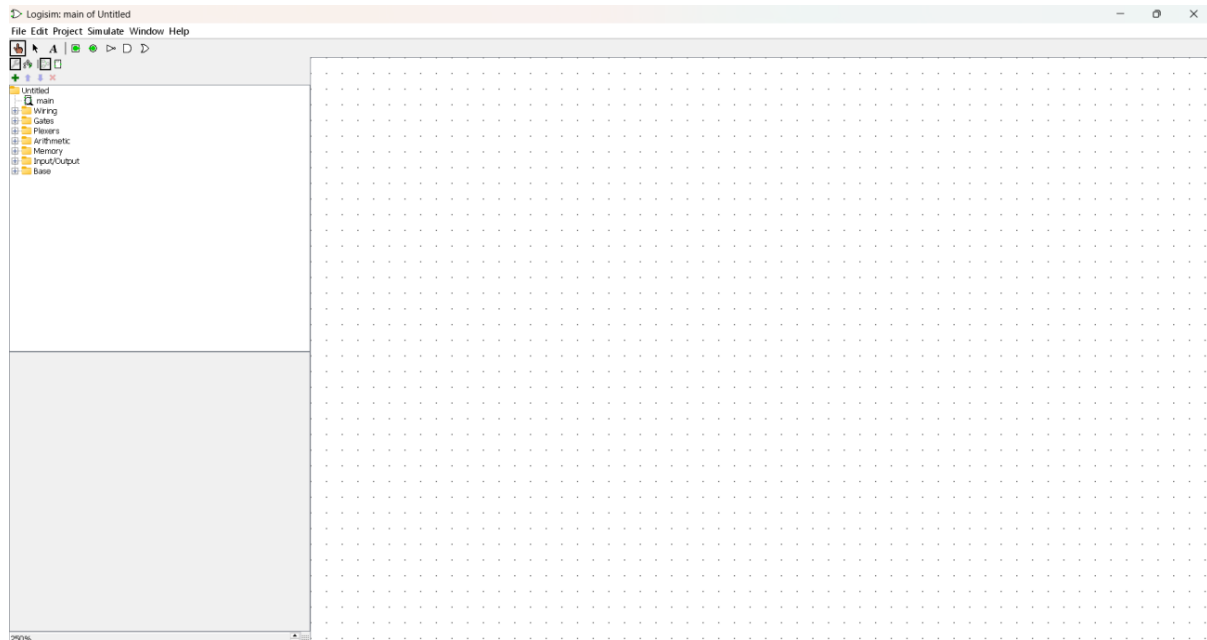
After applying the input and toggling the clock, the parallel output (Q3 Q2 Q1 Q0) = 1 0 1 1.

Procedure:

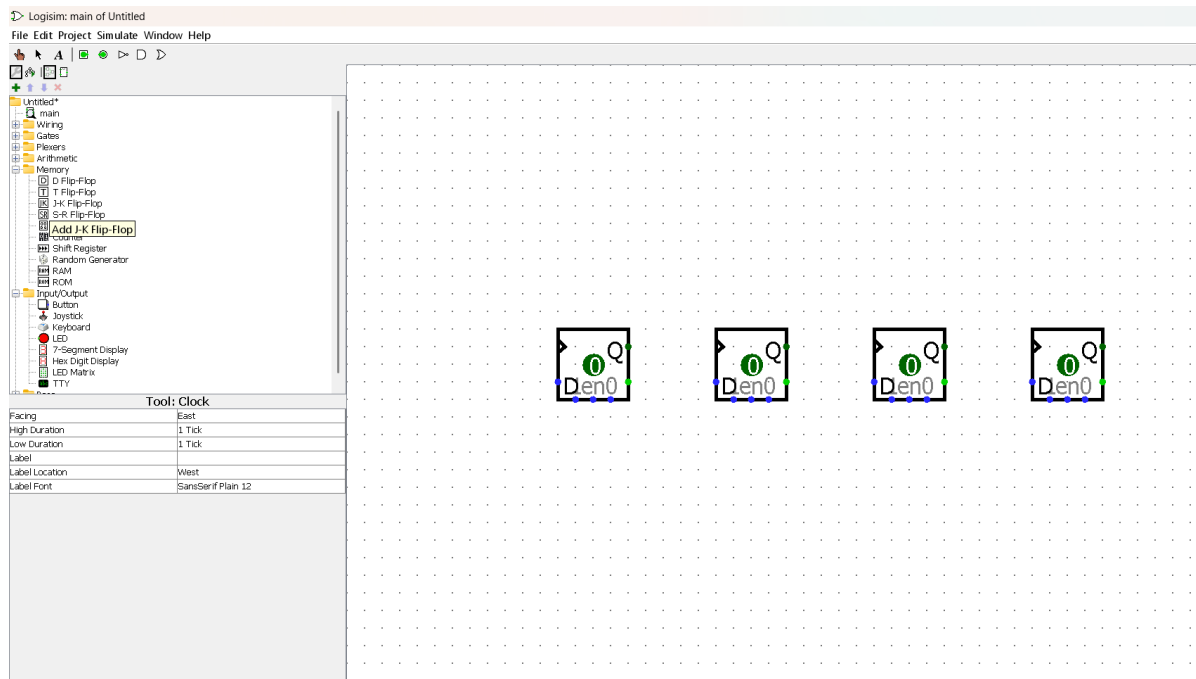
1. Open the Logisim software and select a blank workspace.
2. Place 4 D flip-flops onto the canvas from the memory components menu.
3. Add 4 separate input pins representing the parallel inputs (D3, D2, D1, D0).
4. Connect each input pin directly to the D input of a corresponding D flip-flop.
5. Connect a common clock signal to the clock (CLK) inputs of all four D flip-flops to ensure simultaneous loading of the bits.
6. Connect the clear/reset input of all D flip-flops to a single reset pin to initialize the register to 0000 before the start of the simulation.
7. Add 4 LED indicators and connect them to the Q outputs of each D flip-flop to visually monitor the parallel outputs.
8. Apply the parallel input sequence 1011 to the input pins by toggling their values.
9. Toggle the common clock pulse and observe that the parallel outputs immediately display the applied inputs without shifting.
10. Verify the operation by changing the parallel inputs (e.g., to 0110) and observing the outputs after the next clock pulse.

Simulation Images:

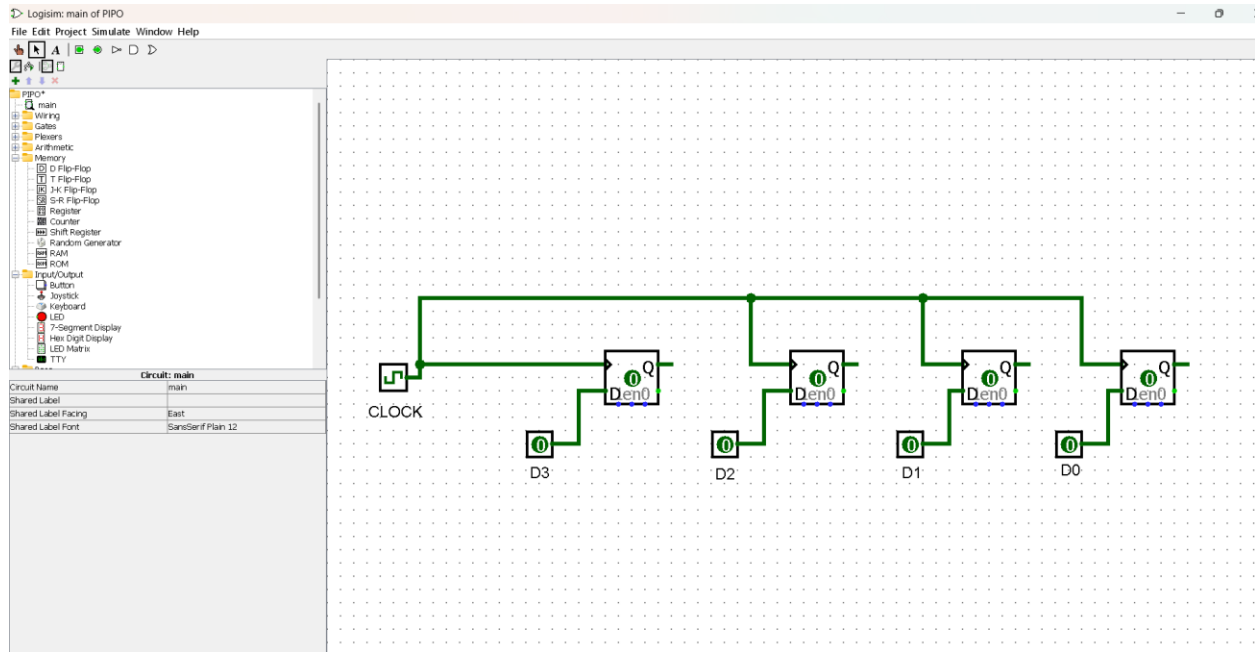
[Step 1: Open Blank Logisim workspace]



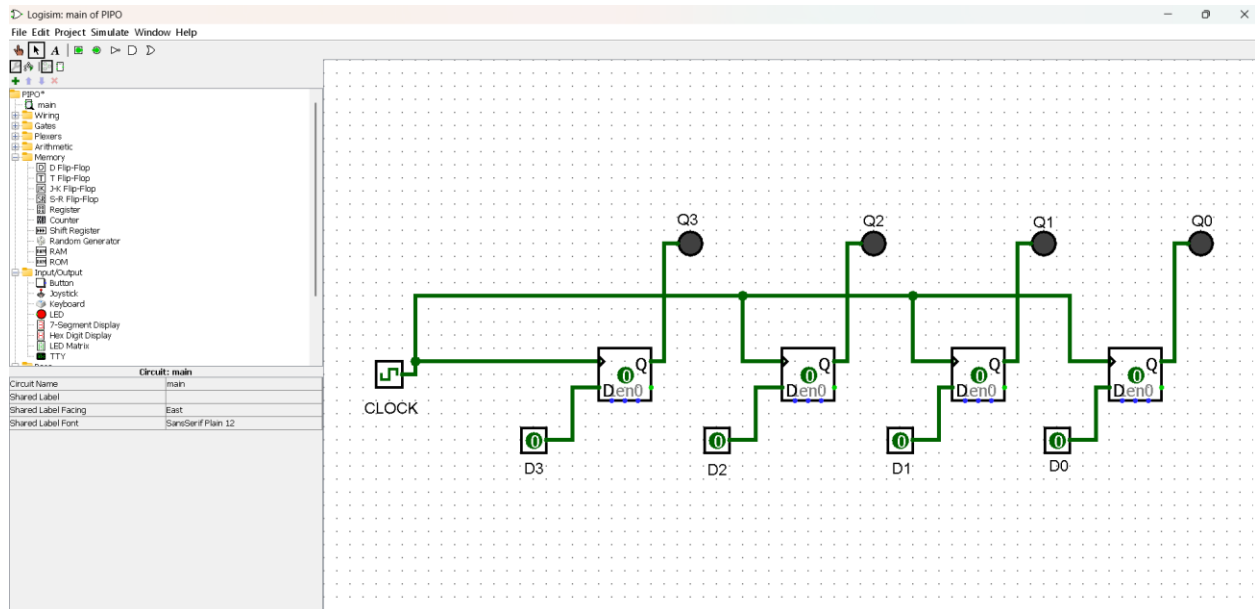
[Step2:Place 4 D Flip-Flops placed onto the canvas]



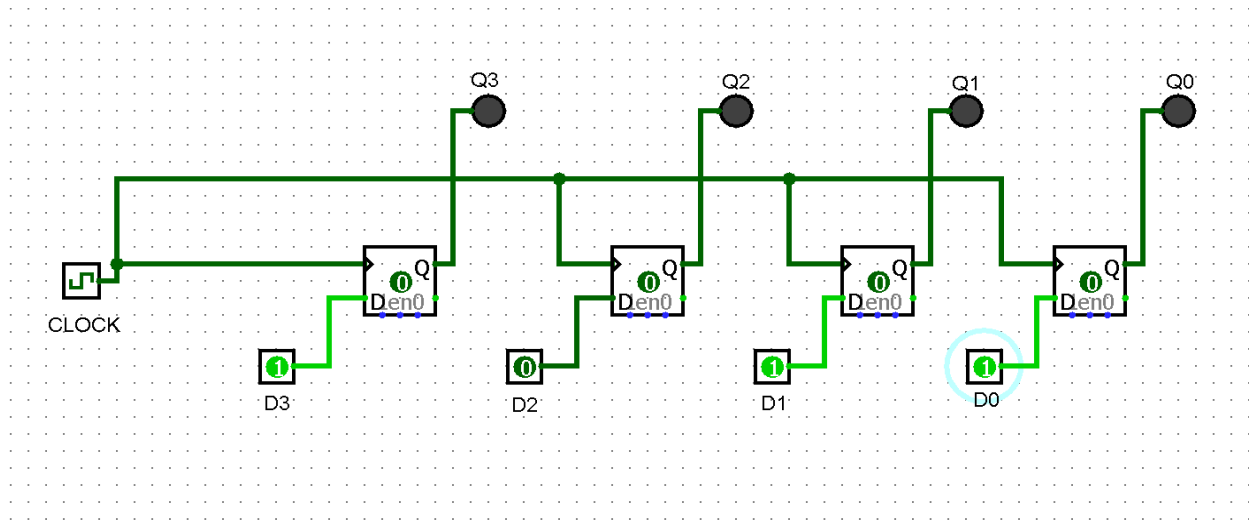
[Step3: Wiring completed (Parallel Inputs, Common Clock, Reset, and Outputs connected)]



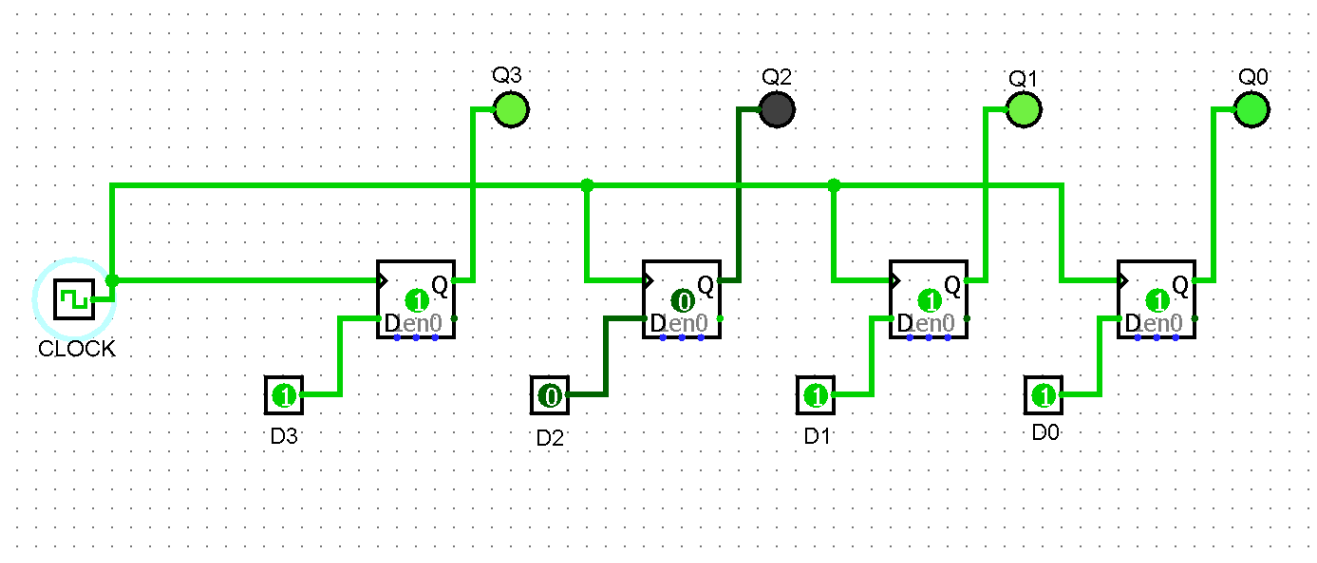
[Step4: LEDs connected to Q outputs]



[Step 5: Parallel input 1011 applied (Outputs remain 0000 before active clock edge)]

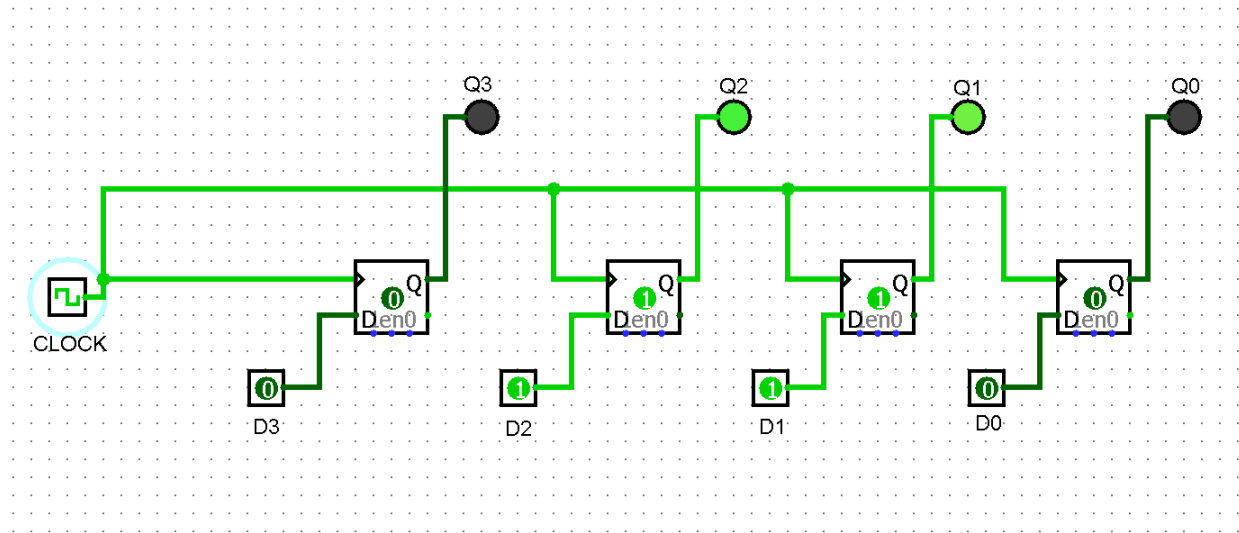


[Step 6: Final output 1011 successfully loaded onto the LEDs after active clock pulse]



Observations:

1. Before clock pulse: The parallel input applied is 1011, but the outputs remain 0000 as the clock has not been triggered to load the data.
2. After applying clock pulse: The output LEDs instantly change to 1011, exactly matching the parallel inputs across all 4 bits simultaneously.
3. Subsequent clock pulses: If the input remains 1011, the output remains unchanged at 1011. If the input is changed (e.g., to 0110), the output instantly updates to 0110 upon the next clock pulse.



Result:

The 4-bit Parallel-In Parallel-Out (PIPO) shift register was successfully simulated using D flip-flops in Logisim. The parallel input sequence 1011 was applied simultaneously across the input pins, and the corresponding parallel outputs were correctly observed instantly after a single clock pulse. The obtained outputs fully matched the expected immediate data-loading operation of a PIPO shift register.

Inference:

The simulation demonstrated that a PIPO shift register loads parallel data simultaneously into all its flip-flops on the active edge of the clock pulse. Unlike serial shift registers which require

multiple clock cycles (one for each bit) to shift data sequentially through the cascaded flip-flops, the PIPO register operates in a single clock cycle. This confirms that there is no bit-shifting delay, making PIPO registers highly efficient for rapid data storage and parallel data retrieval in digital circuitry.

Conclusion:

The objective of simulating a 4-bit PIPO shift register was achieved successfully. The parallel input sequence 1011 was successfully stored in the four D flip-flops with a single clock pulse, producing the expected parallel outputs simultaneously. The experiment successfully validated the working principle of a PIPO register and highlighted its primary application in rapid data storage, buffers, and parallel data transfer within digital systems.